Penn Engineering
University of Pennsylvania

FA26

Special Seminar on
Multi-Modal AI

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Department of CIS, GRASP

https://cis7000-fall26.thelabone.org/

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Lecture 15: Fusion Architectures and Unified Multimodal Models

Special Seminar on Multi-Modal AI

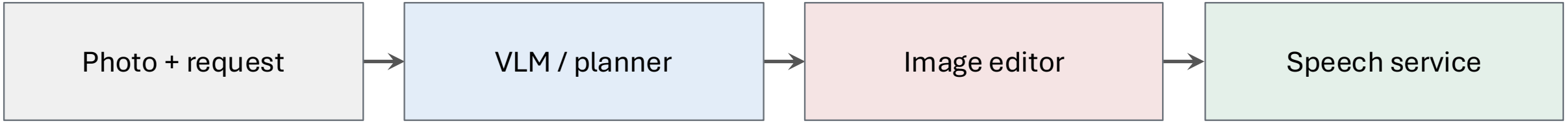
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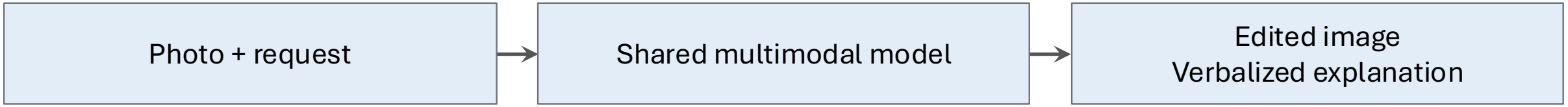
Multimodal Requests

“Move the red mug to the right of the blue plate, then explain the edit aloud”

Software engineering solution:



Alternative:



Goal of This Course

Use Machine Learning to Build Unified Models

Why unify multimodal models?

1 How does unification simplify "deployment"?

Reuse one model interface and its context across tasks

2 How does joint training create cross-task synergy?

Let one task improve another through shared learning

1 · Simplified deployment

How can one multimodal model replace a pipeline of task-specific models?

Recap: Multimodal Understanding

Task	Input and example request	Useful output
Visual question answering	Photo + “Which object is left of the plate?”	Text
Image generation	“Generate an illustration of …”	Image
OCR / document / chart QA	Page + “Read the total and explain this trend”	Text / Json
Video understanding	Clip + “What happened before the door opened?”	Text
Speech / audio understanding	Recording + “Transcribe this”	Text

Each query has a specialized “input-output” interface

Multimodal generation: beyond answering in text

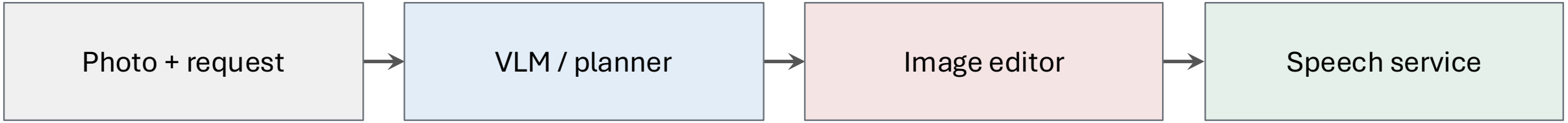
Request	**Input → output**
Generalized image generation	Text → image, or interleaved text + images
Image editing	Source image + instruction → edited image
Video continuation	Frame history + text → subsequent images
Audio QA	Image / video / audio context → text + speech

We will see how to handle these “multimodal” queries

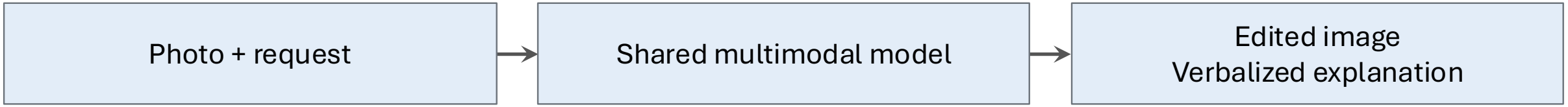
Multimodal Requests

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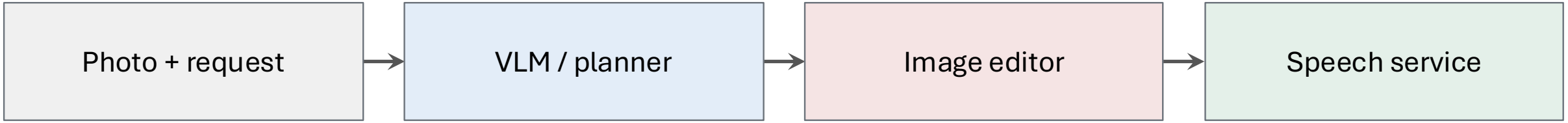
Alternative:



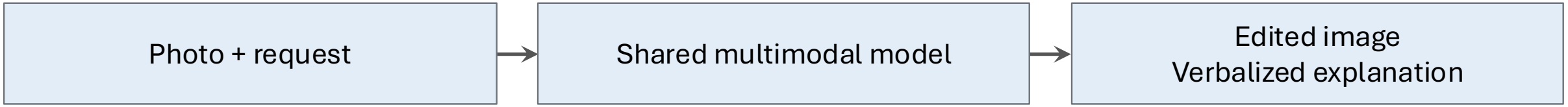
Multimodal Requests

“Move the red mug to the right of the blue plate, then explain the edit aloud”

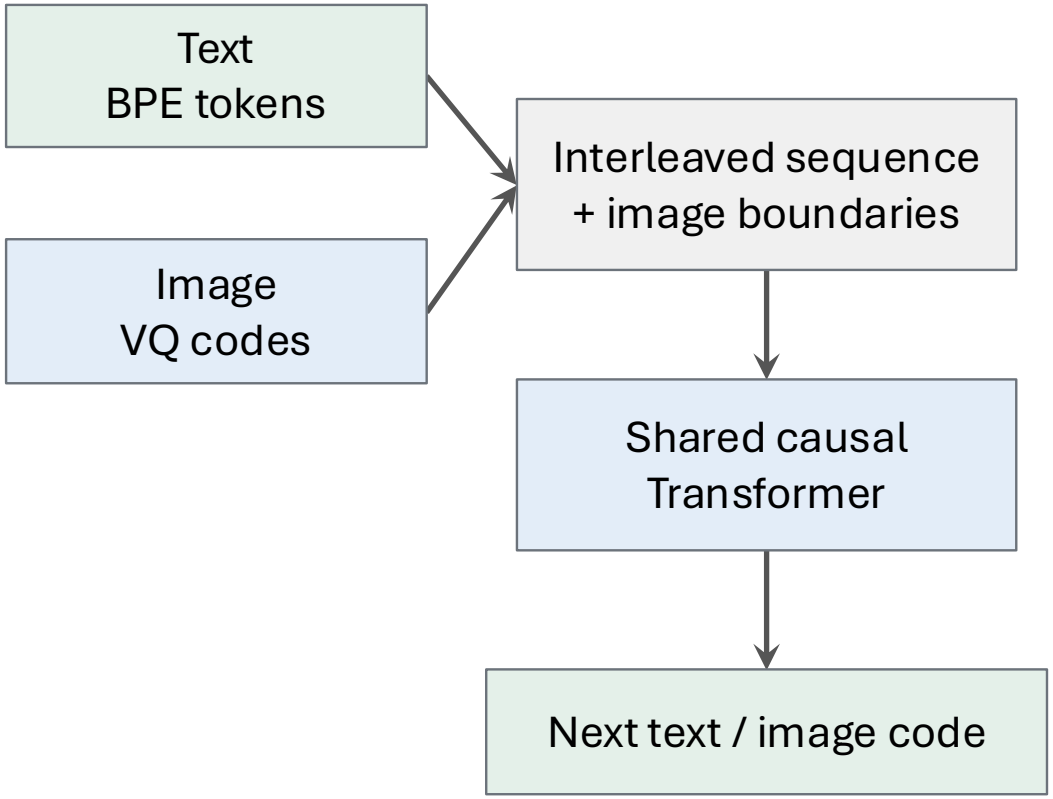
Software engineering solution:

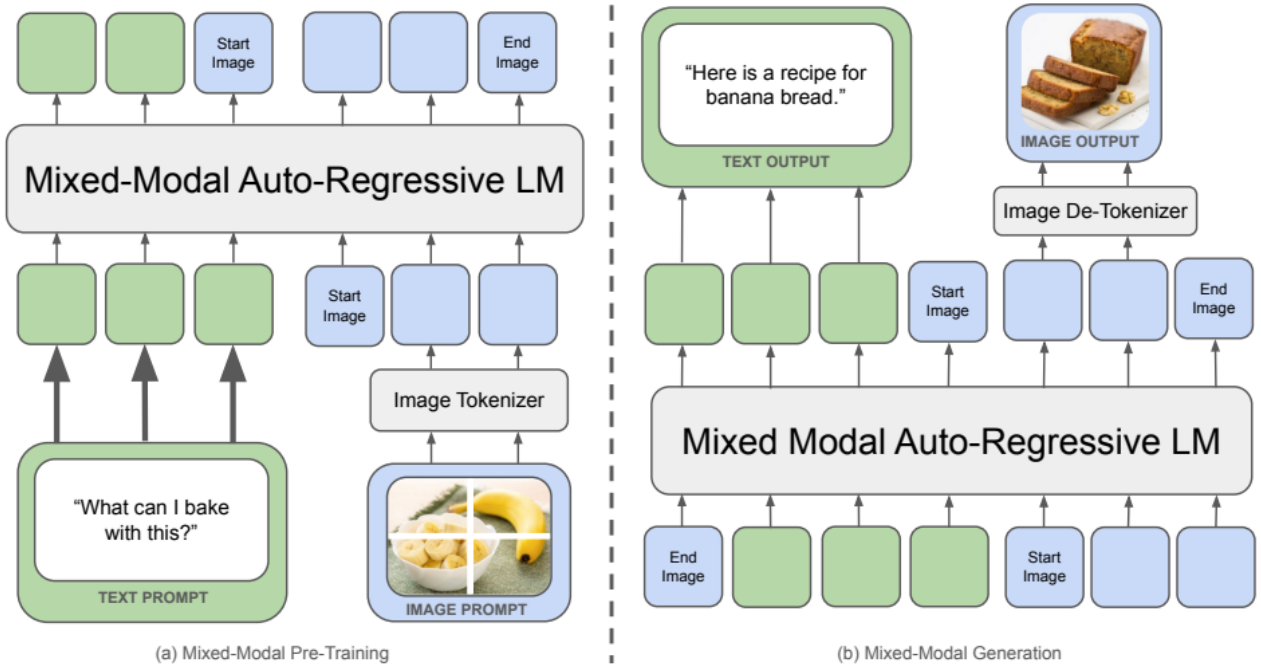


Alternative:



Simplest Idea: Chameleon

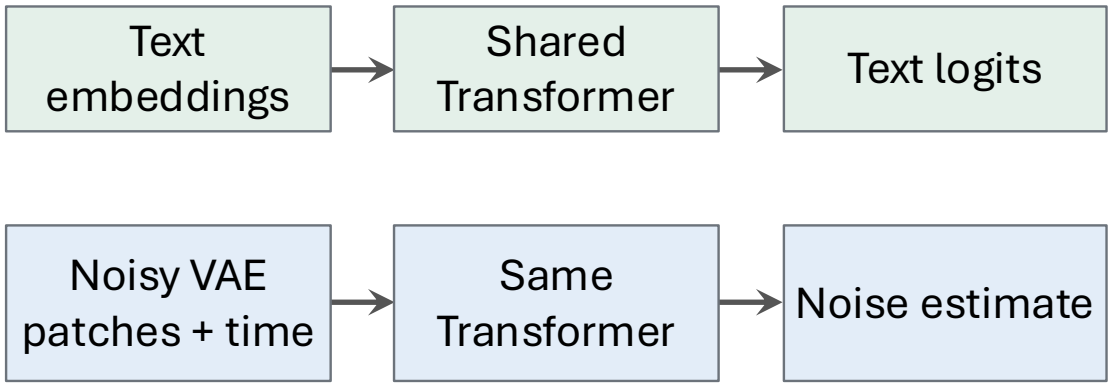


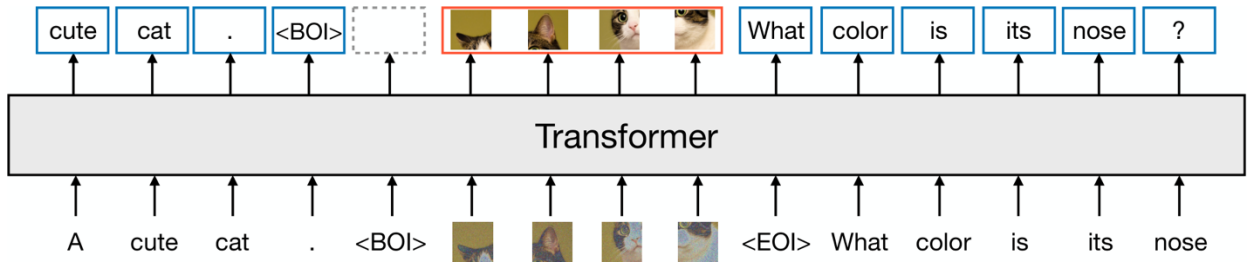


Original: Chameleon, Fig 1

Key idea: a common prediction interface supports mixed text–image documents
Beautiful idea, but transforming images into discrete tokens makes learning hard…

Transfusion: shared trunk, two output distributions





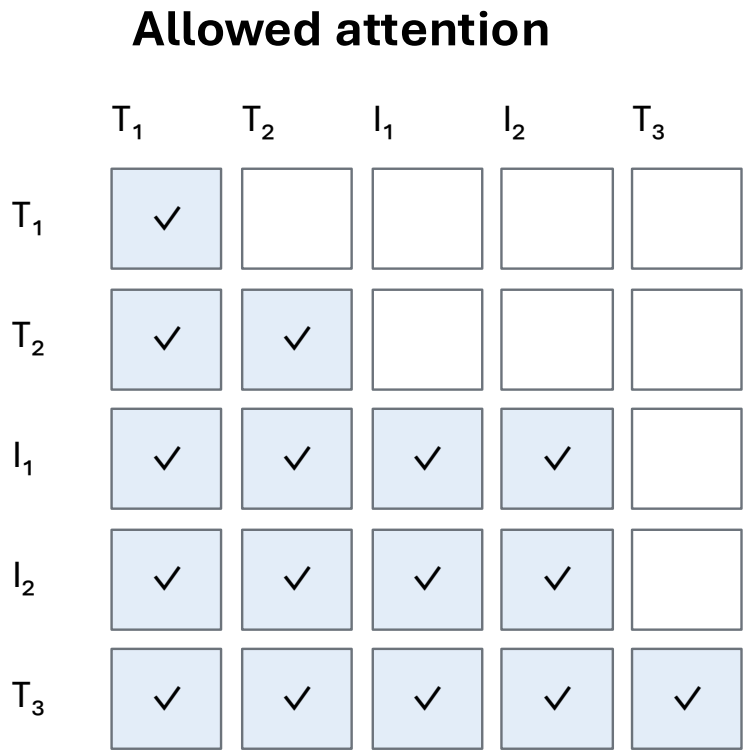
Shared weights receive both losses

Image sampling uses the diffusion loss

Key idea: unifying parameters does keep two “decoding heads”
From the same embeddings, use MLP1 to predict text logits, MLP2 to predict image noises

Transfusion: Playing with the Attention Mask

Text: causal, next-token-prediction loss
Image block: bidirectional, diffusion loss



Training data

Learning problem	Training sample	Reported sources / construction
QA	Image + question / caption + target text	Web alt-text, OCR, charts, grounding BAGEL understanding data
Generate or edit	Caption → image Source + instruction → target image	Curated image–text pairs, synthetic images OmniEdit and other editing datasets
Use visual history	Interleaved frames / images + descriptions	Koala36M, MVImgNet2.0, online videos OmniCorpus / Common Crawl
Understand and speak	Audio + transcript, sound caption or dialogue	Common Voice, LibriSpeech, WavCaps, AudioCaps TTS-generated questions / conversations

Deng et al. (2025) · §§2–7 / Li et al. (2024) · §IV-B, Tables II, IX / Qwen Team (2025) · §§2–5

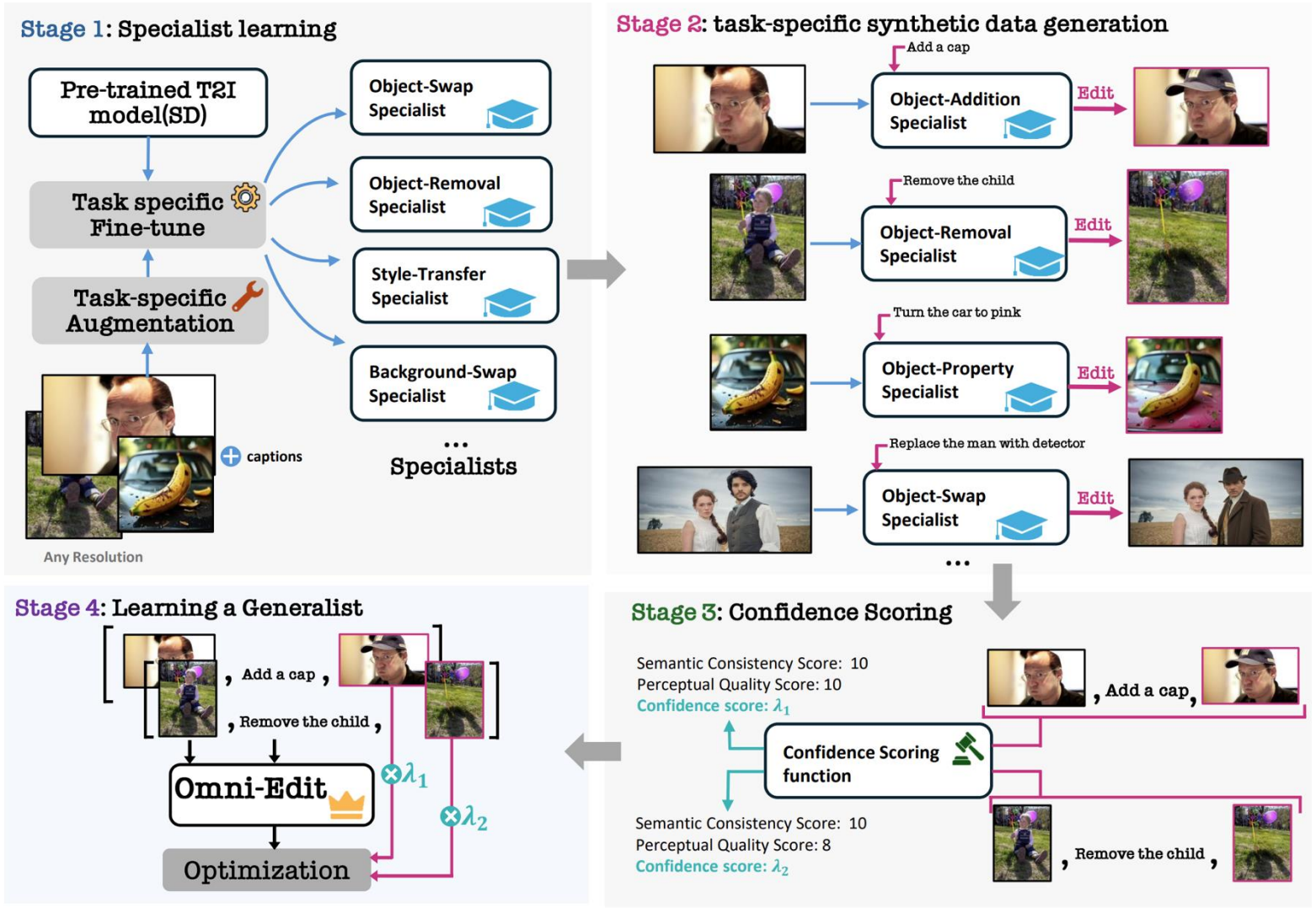
OmniEdit: learning an image edit

Training example = source image + instruction + target image

Source image	Editing instruction	Target image
	Add a cap	
	Remove the child	

Wei et al. (2024) · Official data-construction examples · OmniEdit project / paper

Really Behind the Scene



https://tiger-ai-lab.github.io/OmniEdit/

Video and multiview data

Koala-36M: a clip + detailed caption

video segment
start time … end time

Content: a white car on a rural road
Motion: follows the bends
Camera: stationary, elevated view

MVImgNet2.0: Multiview Images



Examples: bowl and toy car

BAGEL turns sampled frames and change descriptions into interleaved supervision.

Koala-36M project

MVImgNet2.0 project · Official multiview examples

OmniCorpus: an illustrated document

Common Crawl web pages → extracted, filtered documents

Text: introduce a repair

Image: show the part

Text: explain the next step

Image: show the next view

Data Example:

[Orig. Text] Tips to Take Care of Your Car

[Img Cap.] A mechanic in a garage inspects a car engine, holding a tablet and a tool. They wear gloves and a cap, focusing on the task.

[Image]



[Orig. Text] Follow Maintenance Schedule
The easiest way to take care of your car is by following the maintenance schedule .....

[Img Cap.] A gloved hand holds a car's coolant reservoir cap, emphasizing routine maintenance. The yellow and blue glove contrasts with the dark engine bay

[Image]



[Orig. Text] Check Oil & Other Fluids
Your car depends on several oils and fluids such as engine oil, coolant, and brake fluid in order to .....

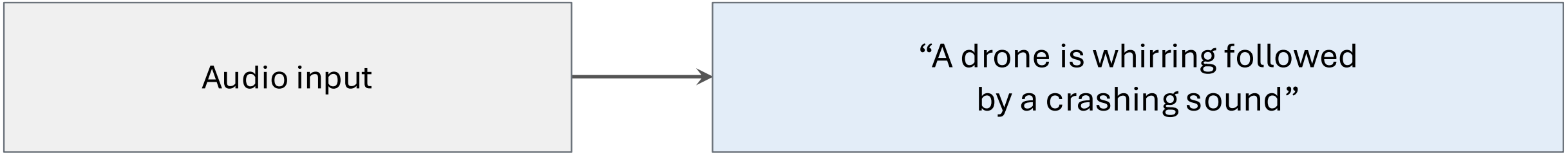
BAGEL paper: processed web example
Original page text + added image captions

Li et al. (2024) · OmniCorpus

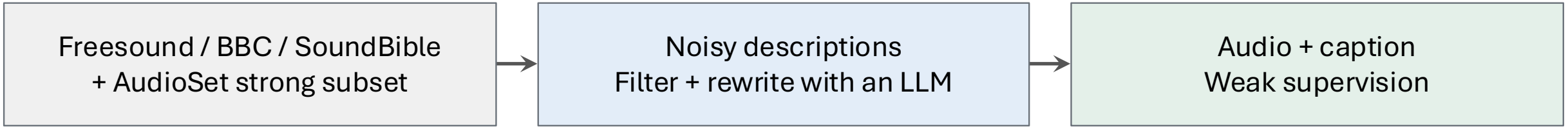
Deng et al. (2025) · BAGEL data example

Audio captions

AudioCaps: audio paired with a human caption



WavCaps: web audio paired with cleaned descriptions



For our mug task: understand the spoken request, then recognize the clink.

AudioCaps (Kim et al., 2019)

WavCaps (Mei et al., 2024)

BAGEL: where interleaved supervision comes from

Video: online clips + Koala36M + MVImgNet2.0



Web: OmniCorpus documents from Common Crawl



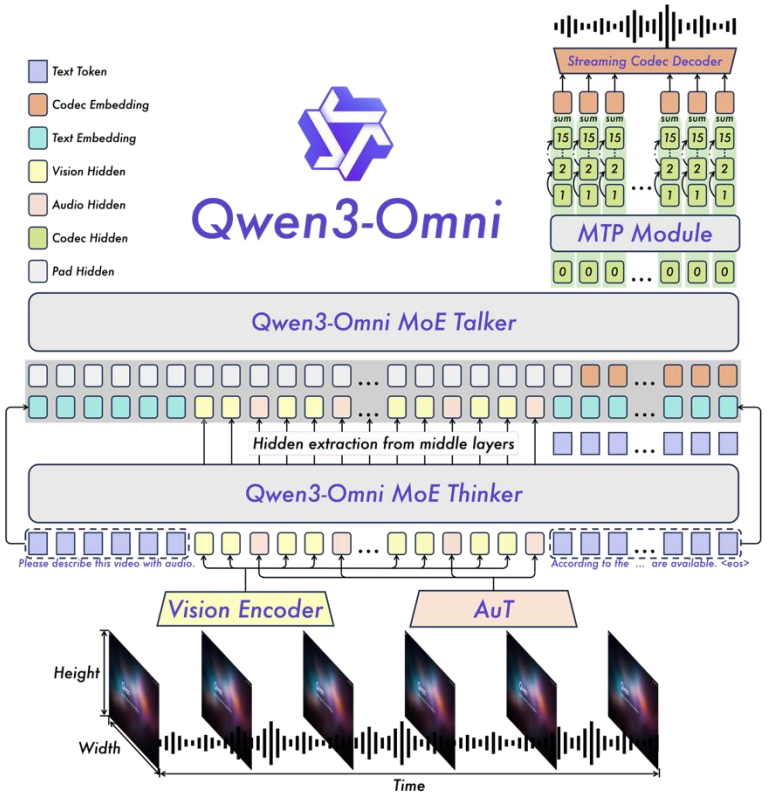
Also: 500M paired-understanding examples + 1,600M paired-generation examples

500k reasoning-augmented examples add guidance before generation / editing

Qwen3-Omni: Thinker and Talker



Module	Architecture	Params	Streaming
Audio Encoder	AuT	650M	✓
Vision Encoder	SigLIP2-So400M	540M	-
Thinker	MoE Transformer	30B-A3B	✓
Talker	MoE Transformer	3B-A0.3B	✓
MTP	Dense Transformer	80M	✓
Code2wav	ConvNet	200M	✓
End-to-End First-Packet Latency: **234/547ms**			



Original: Qwen3-Omni, Fig 2

Key idea: think about audio models as “renderers” on tops of the basic “language Transformer”

Unified Models for Multimodal Tasks

The mechanism

Reuse the backbone, context and task interface
Reduce separately maintained models

Engineering challenges

Encoders, output heads and generation schedules (e.g., real-time audio streaming)

Additional Thoughts

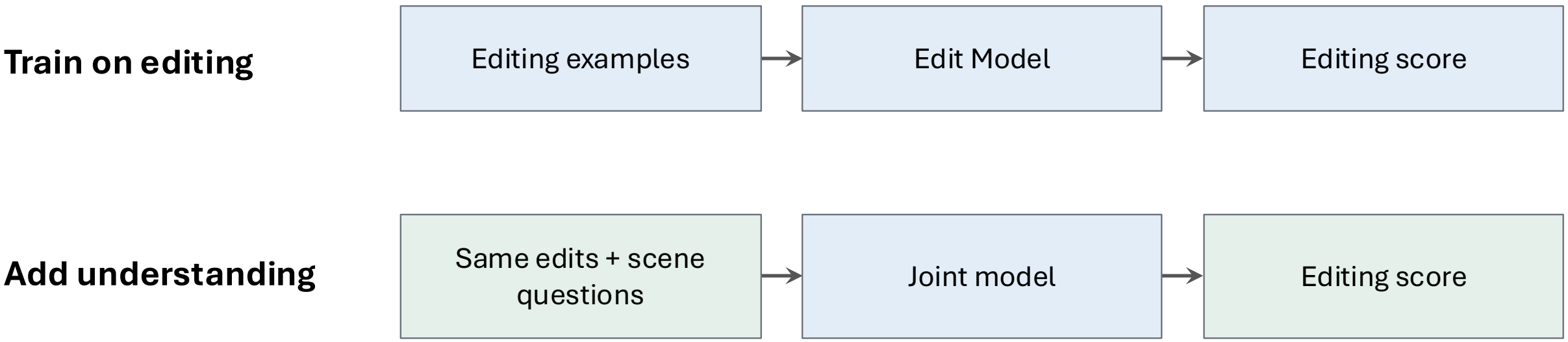
The “recursive-self-improvement” has been happening, for long

CLIP + segmentation gives “promptable segmentation”
CLIP is largely used in data cleaning processes
VLMs + image edit models gives “reasoning-augmented” image editing

2 · Cross-task synergy

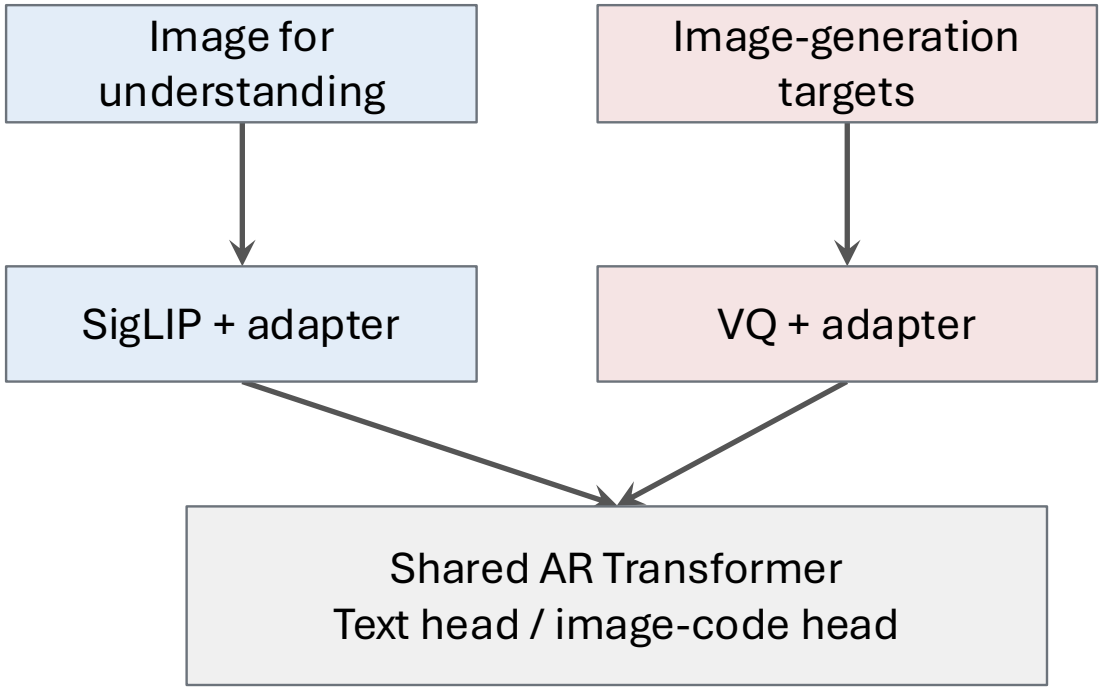
When does learning one task make another task better?

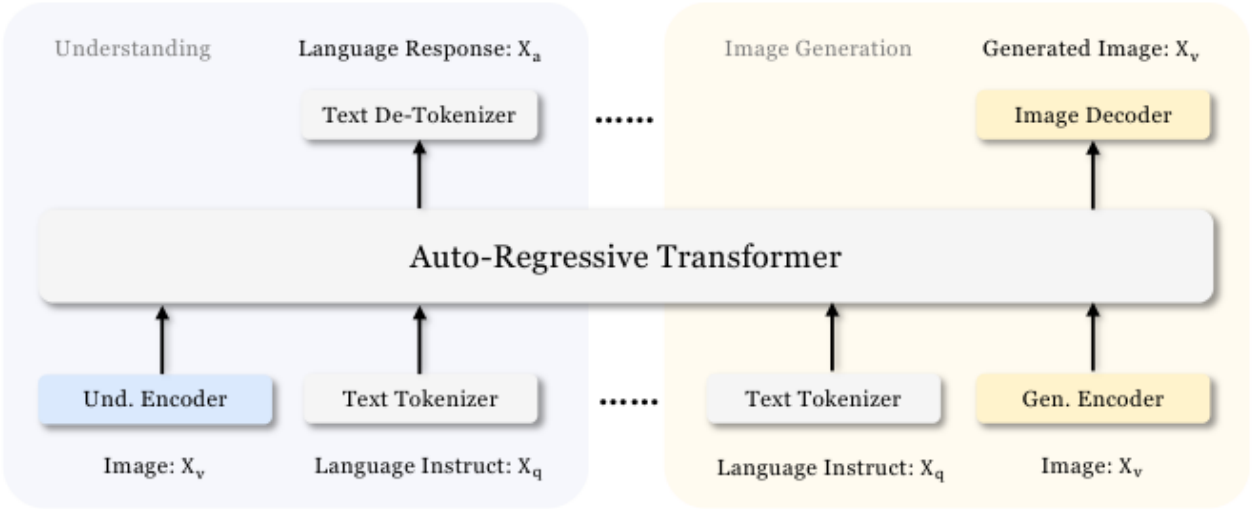
Example: understanding should improve editing



Issue: editing task requires image features to contain details…
Fundamentally, we need “task-specific” visual features

Janus: separate visual encoders, shared generator



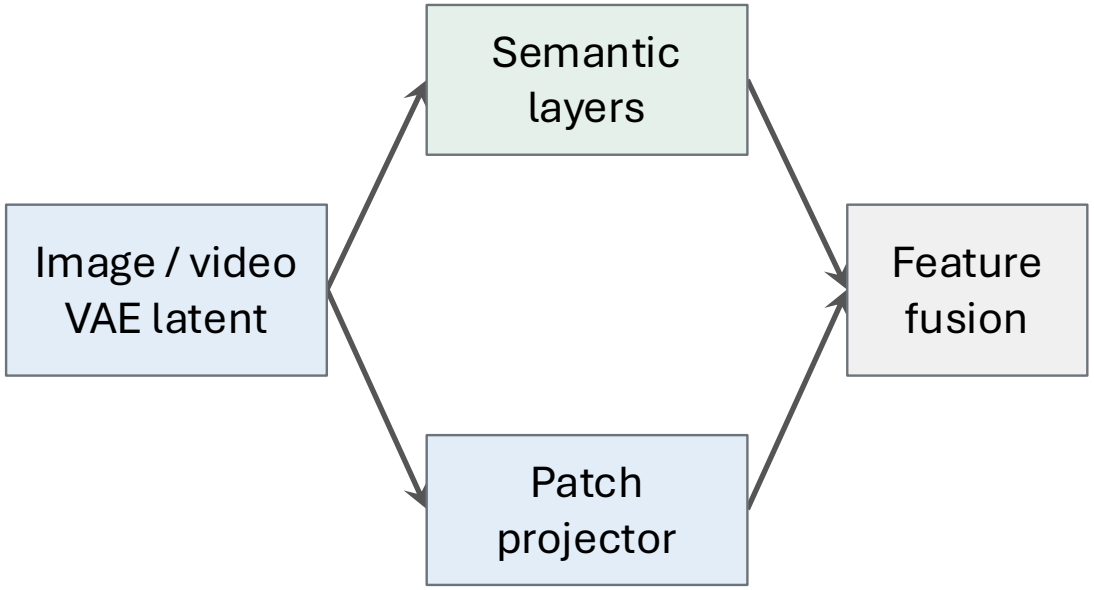


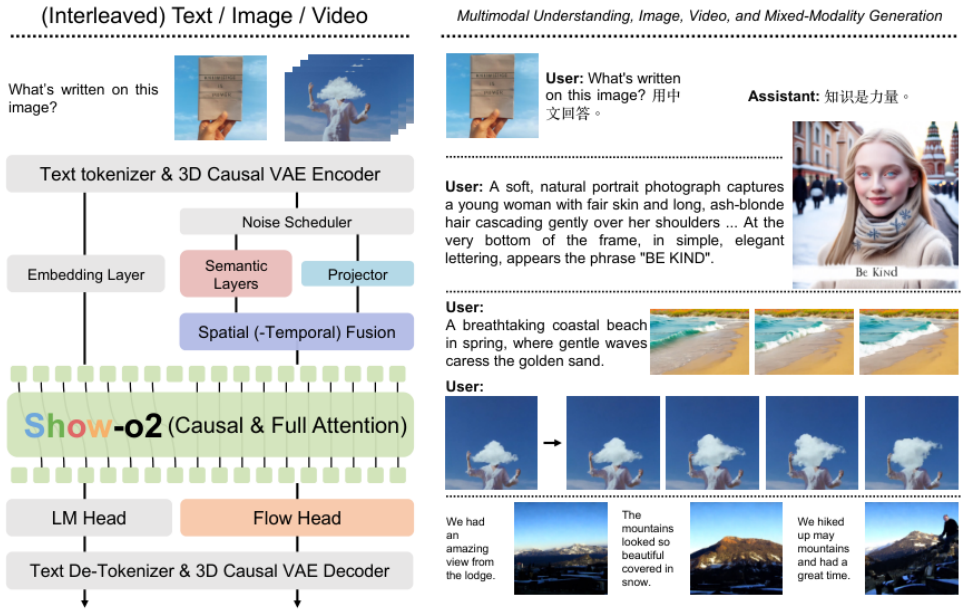
Original: Janus, Fig 2

The basic architecture is Chameleon

Benefit: task-appropriate features **Limit:** need to align the interface

Show-o2: fuse semantic and low-level features





Original: Show-o2, Fig 1

Shared LLM + language head + flow head

$$\underbrace{S(z_t)}_{\text{features from VAE latents}} \approx \underbrace{\text{SigLIP}(I)}_{\text{teacher features from the image}} .$$

The basic architecture is Transfusion

Benefit: task-appropriate features **Limit:** need to align the interface

Xie et al. (2025) · §3, Fig 1

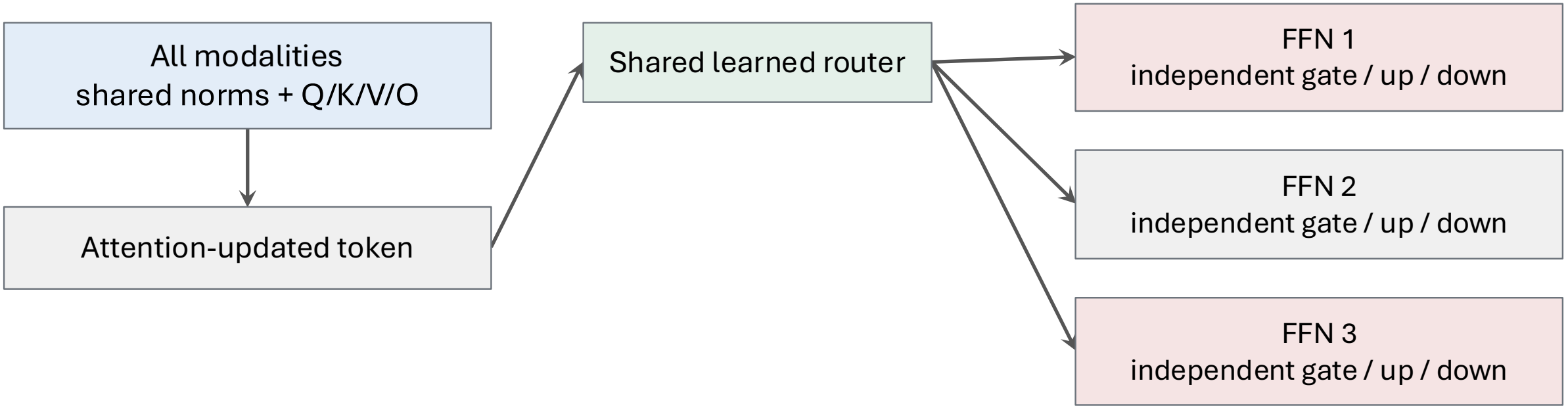
Task synergy from an gradient perspective

$$g_A = \nabla \mathcal{L}_A, \qquad g_B = \nabla \mathcal{L}_B$$

$$\mathcal{L}_B(\theta - \eta g_A) - \mathcal{L}_B(\theta) \approx -\eta\, g_A \cdot g_B$$

Intuition: $cos\ \varphi = \frac{g_A \cdot g_B}{\|g_A\|\,\|g_B\|}$

MoE: shared attention, selectively used FFNs

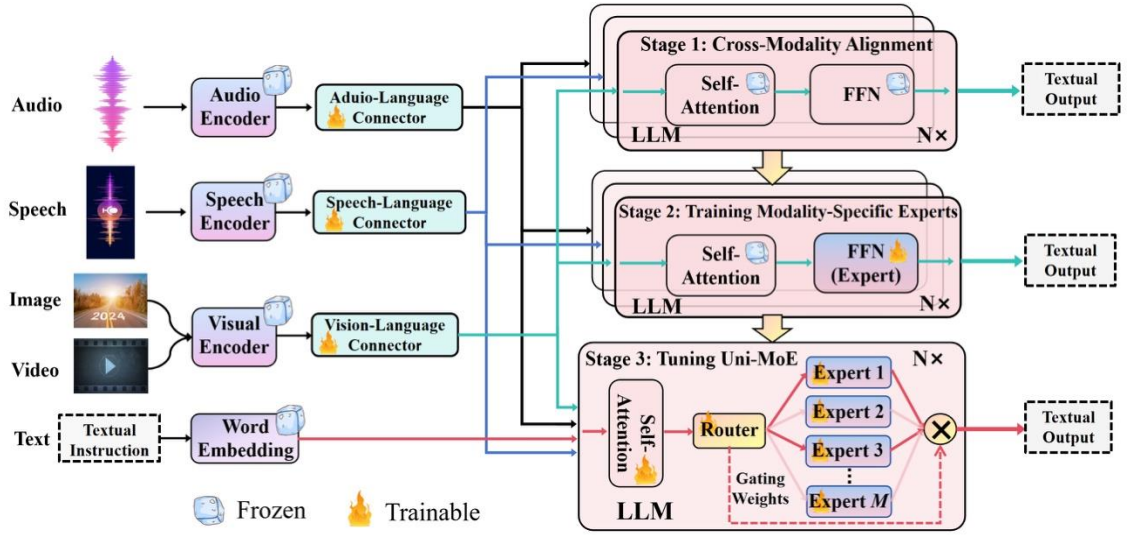


$$g(h) = softmax(W_r h), \qquad S = TopK(g(h))$$

$$y = \sum_{i \in S} \frac{g_i}{\sum_{j \in S} g_j} E_i(h)$$

Uni-MoE: initialize expertise before learning routing

Align modality connectors → Pretrain expert preferences → Mixed-data routing + LoRA



Key Idea: Give experts different training histories before mixing them

Original: Uni-MoE, Fig 2

Uni-MoE: more image data helps video and speech

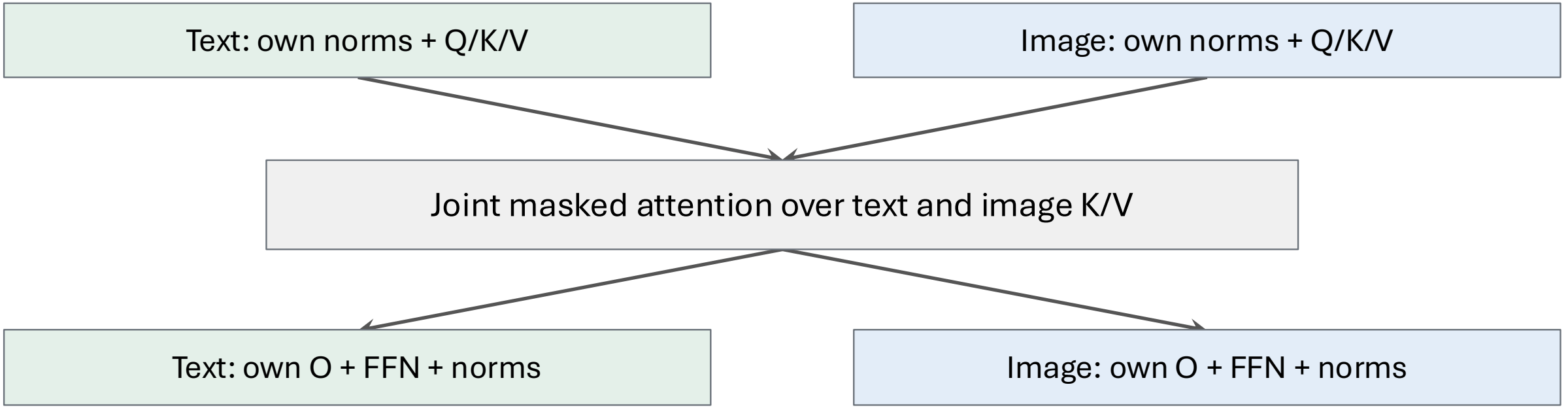
Same 4 experts, top-2 routing: LLaVA-150k becomes LLaVA-v1.5-665k

Held-out task	MoE-Task2	MoE-Task4	Change
ActivityNet-QA accuracy	42.2%	45.6%	+3.4 pp
RACE-Audio: middle	49.65%	64.21%	+14.56 pp
RACE-Audio: high	49.37%	64.72%	+15.35 pp

Conclusion: “better” image-text data helps other tasks; Caveats: still more data, and “text-training”

Li et al. (2024) · §IV-B, Tables II, IX

MoT: independent projections, joint attention



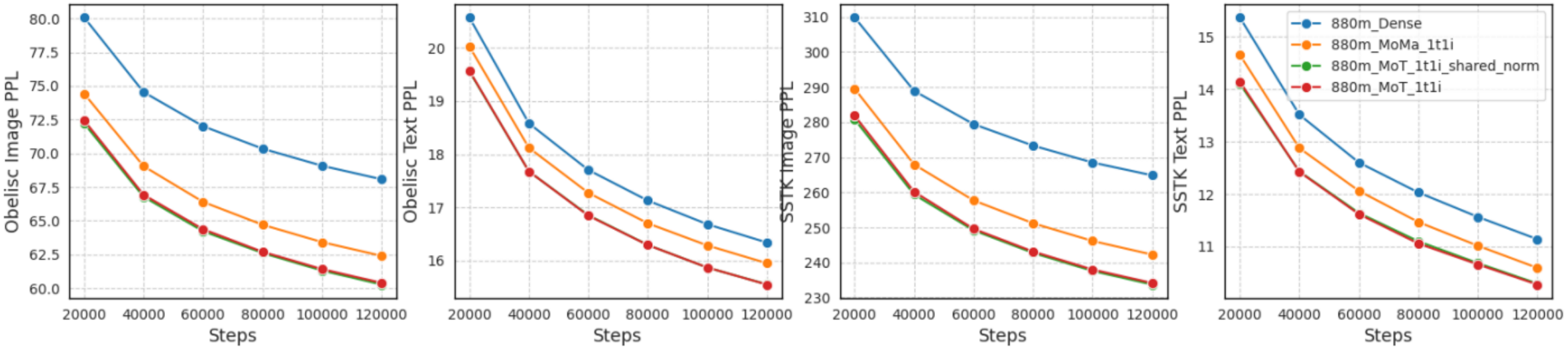
$$q_i = x_i {W_Q}^{m_i}, \qquad k_i = x_i {W_K}^{m_i}, \qquad v_i = x_i {W_V}^{m_i}$$

$$a_{ij} = softmax_{j \in C(i)}(q_i {k_j}^T / \sqrt{d}), \qquad o_i = \sum_{j \in C(i)} a_{ij} v_j \, {W_O}^{m_i}$$

Fixed token-type route Same-width forward arithmetic stays approximately dense

Liang et al. (2025) · §2, Fig 1

MoT: which components benefit from untying?



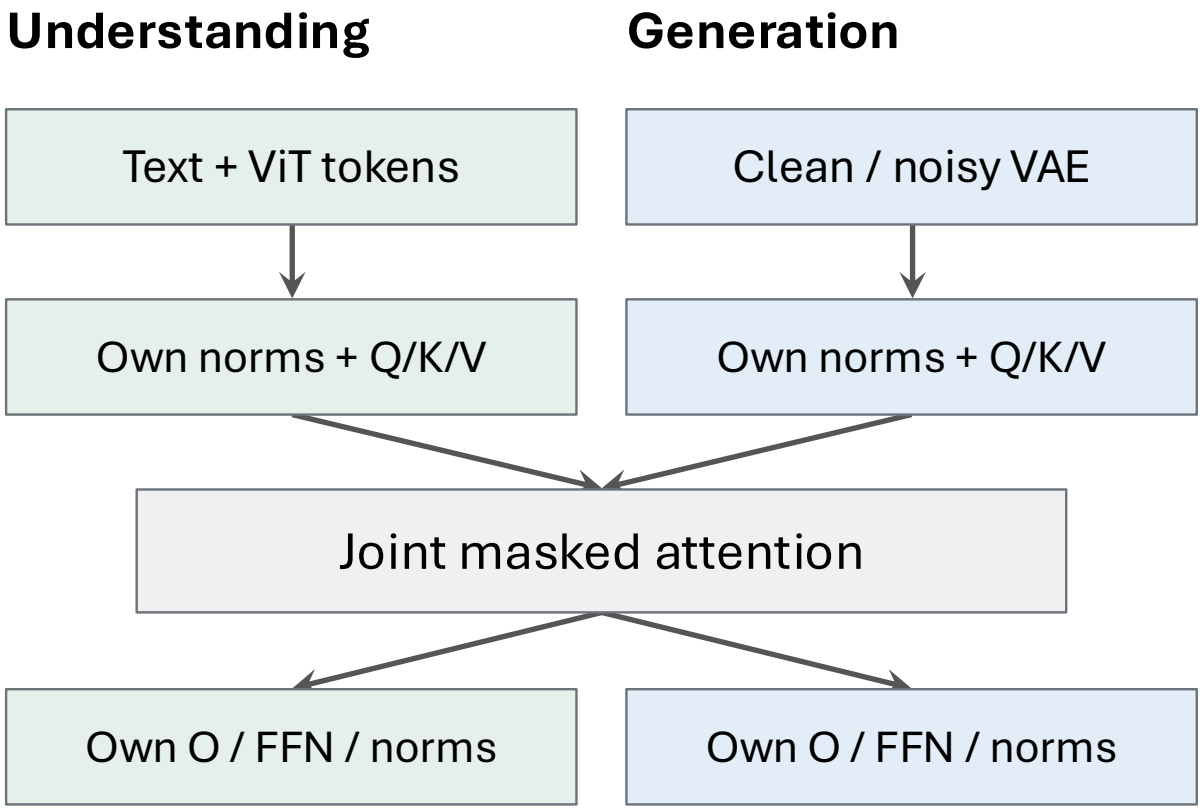
Original: MoT, Fig 14 · 880M active scale · Chameleon setting · held-out perplexity

Additional untying	Reported result at matched quality
Attention + FFN	≈33.3% fewer image FLOPs / ≈10% fewer text FLOPs
Norms + FFN + attention	Negligible further gain in this setting

Controlled: FLOPs; Not Controlled: #Parameters

Liang et al. (2025) · §3.5, Fig 14

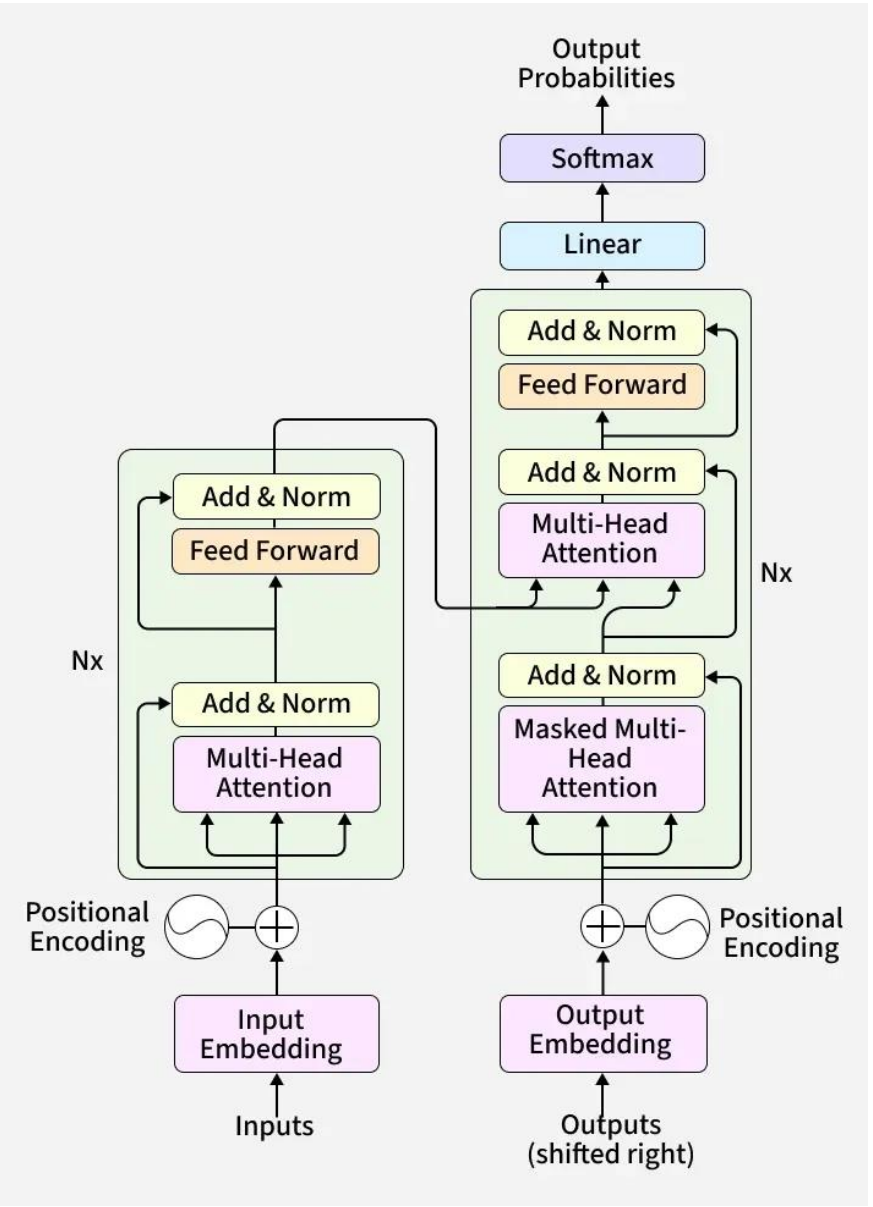
BAGEL: the route follows the toke





Original: BAGEL, Fig

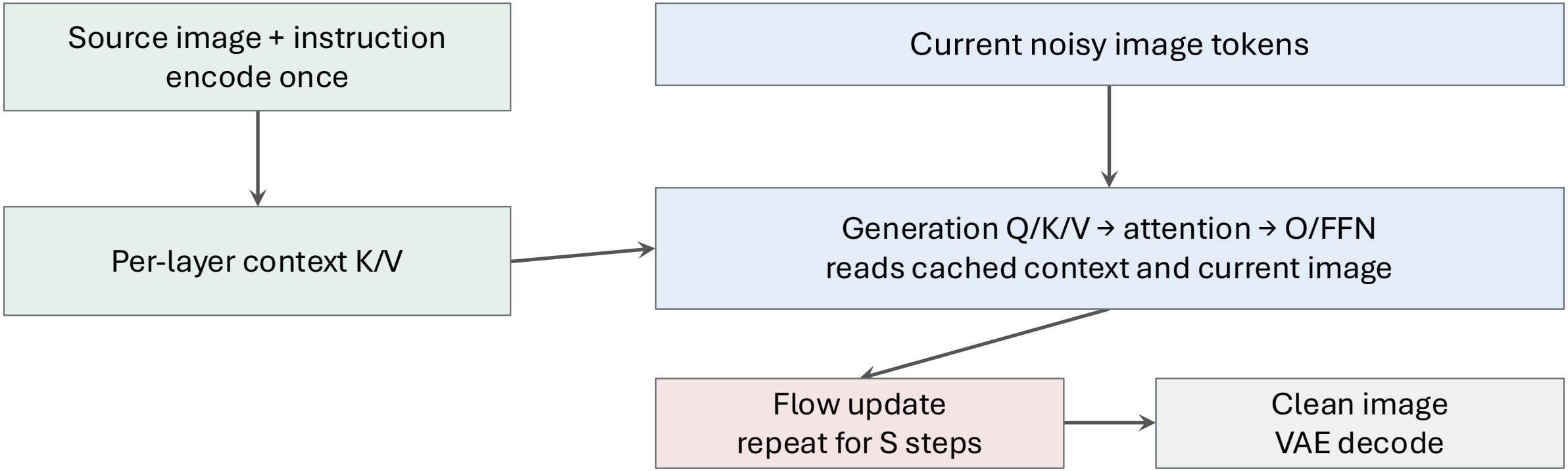
Same init
Independ



BAGEL is “task-driven” MoT

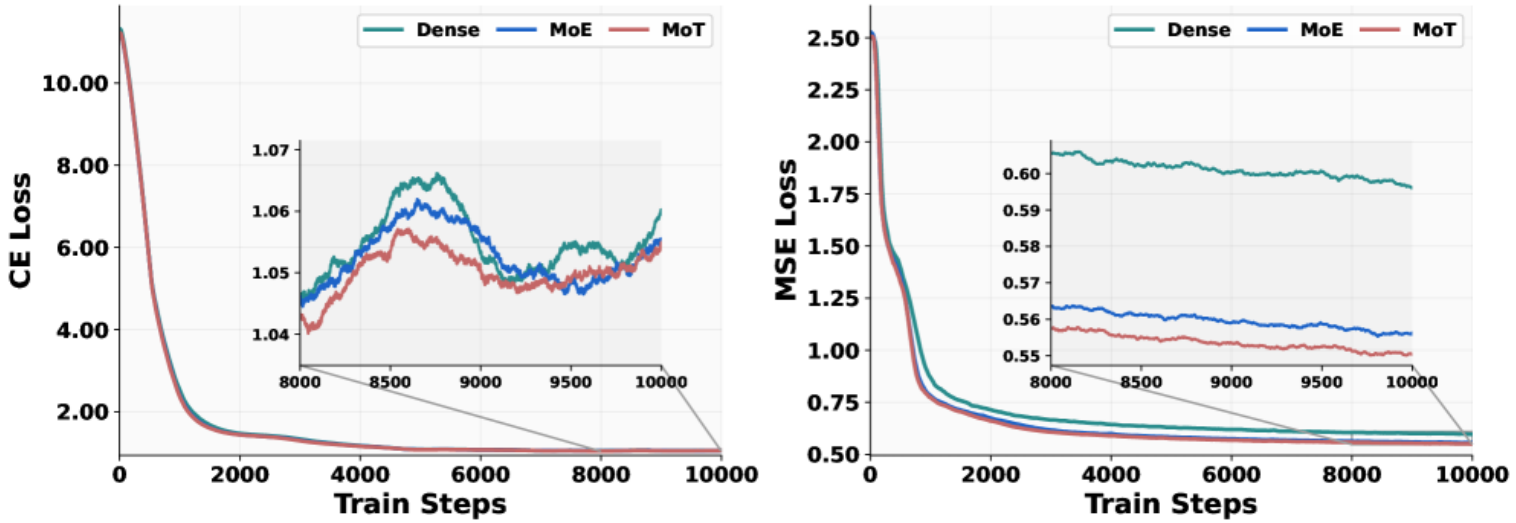
Deng et al. (2025) · §§2–4, Figs 2–3

BAGEL deployment: caching and denoising



Deng et al. (2025) · §§2–4, Figs 2–3

BAGEL: parameter separation at fixed compute



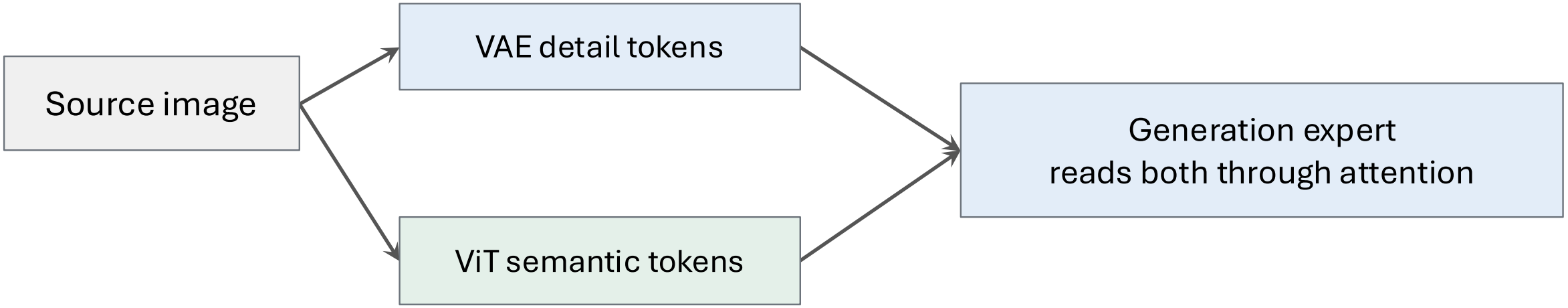
Dense: all weights tied
Hard MoE: FFNs untied
MoT: full pathway untied

Same data, settings and FLOPs
More total parameters
MoT: best generation-loss curve

Original: BAGEL, Fig 3 · Qwen2.5-1.5B initialization · generation:understanding = 4:1

Controlled: FLOPs; Not controlled: #Parameters

BAGEL: semantic features help



Remove ViT context	**Peformance**
Conventional image editing: GEdit	Little change
Intelligent Editing	“16% drop”

BAGEL: semantic features help

	INPUT	INSTRUCTION	DESIRED OUTPUT
Explicit edit	 The requested change is specified.	Make the mug blue. Keep the fox design unchanged. →	
Reasoning-based edit	 Recognize the motif, then adapt it to a new object.	Create a matching tote bag using the design on this mug. →	

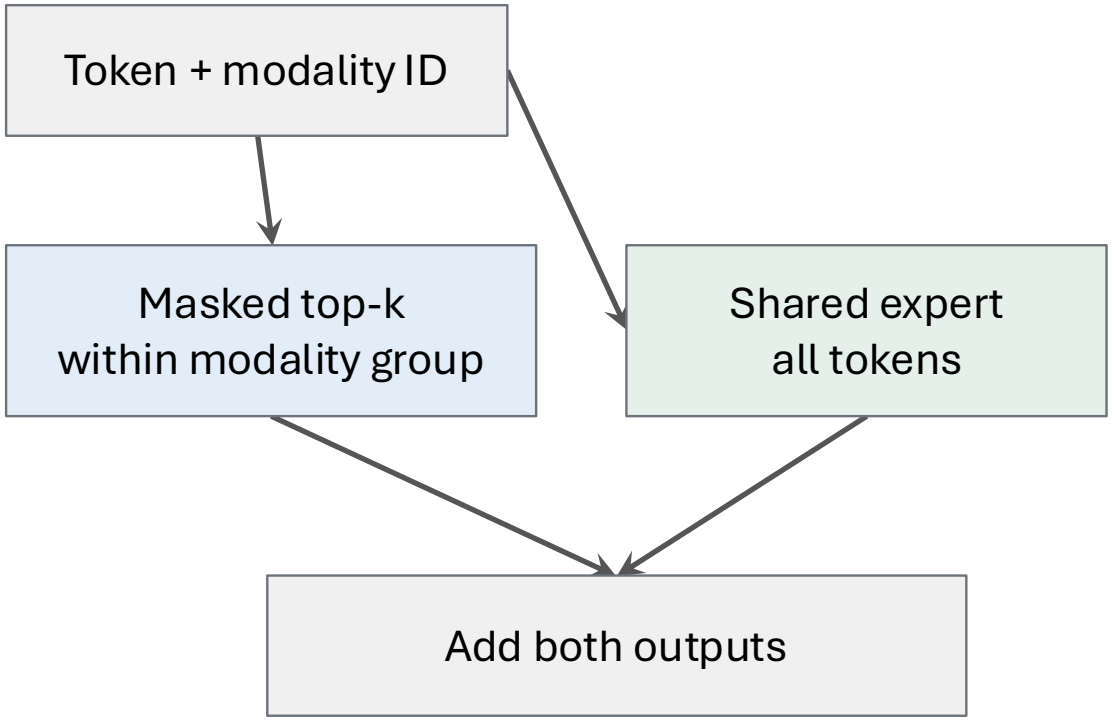
BAGEL: thinking helps

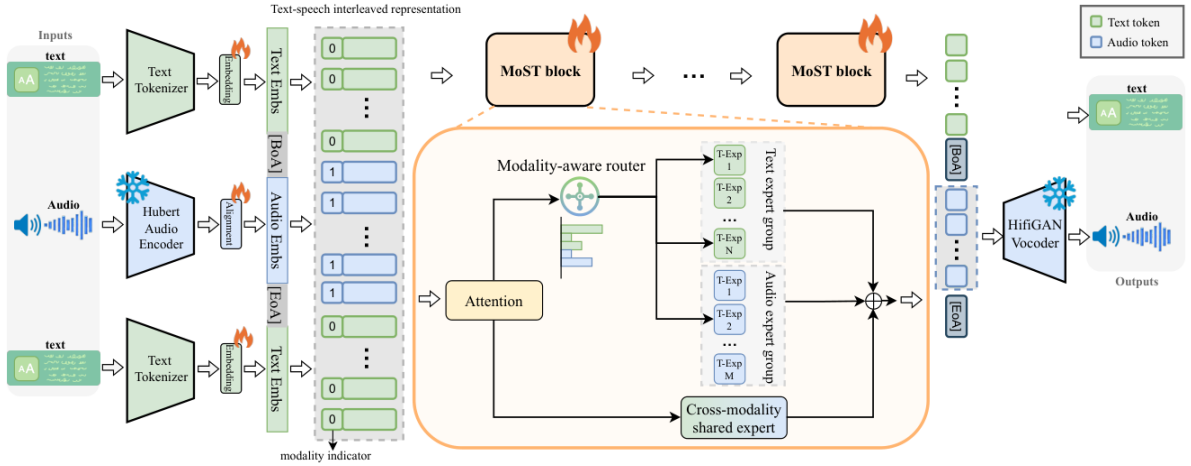
Reported BAGEL model, with or without Self-CoT before visual generation

Benchmark ↑	Without thinking	With Self-CoT	Change
WISE: knowledge-based T2I	0.52	0.70	+0.18
IntelligentBench: editing	44.9	55.3	+10.4
RISEBench: editing	6.1	11.9	+5.8
KRIS-Bench: editing	56.21	60.18	+3.97

Deng et al. (2025) · §§2–7

MoST: modality routing plus an always-on path

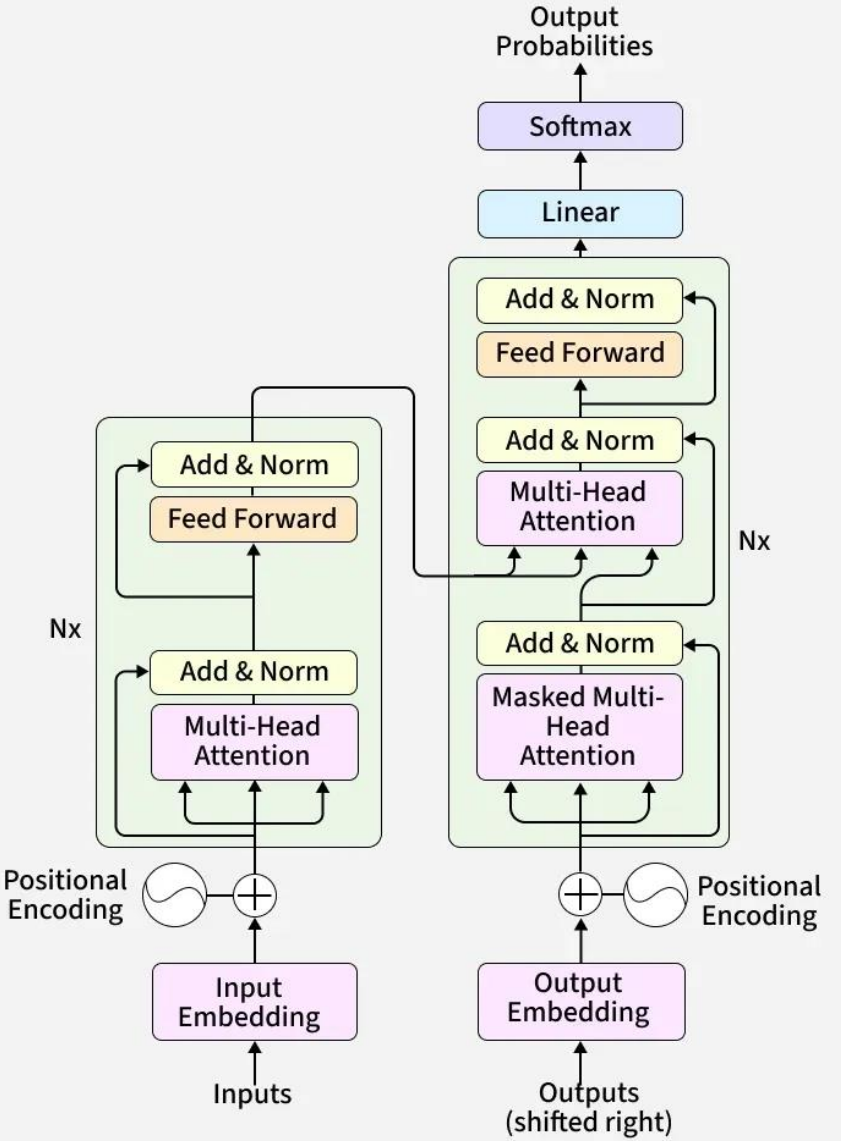




Original: MoST, Fig 1

Parameter sharing

Component	Learned FFN MoE	Modality MoT	BAGEL
Q / K / V / O	Shared across token types	Independent per modality	Independent per U/G route
FFN gate / up / down	Independent expert	Independent per modality	Independent per U/G route
Norm parameters	Usually shared	Independent per modality	Independent per U/G route
Router	Learned top-k	Fixed modality ID	Text + ViT: U; VAE: G



Liang et al. (2025) · §2, Fig 1 / Deng et al. (2025) · §§2–4, Figs 2–3

Synergy

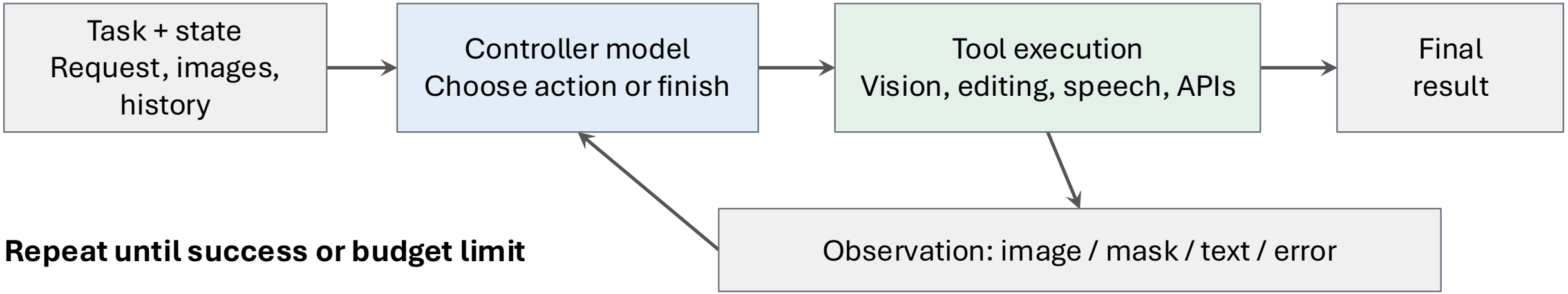
Auxiliary task A	Target task B	Held-out outcome
Image understanding	Image editing	Correct relation?
Image generation / editing	Image understanding	Spatial reasoning?
Speech supervision	Text / visual response	Answer quality?

It is generally agreed that understanding helps (hard) generation, and CoT helps

Questions for Discussion

Agent systems and tool use

A controller selects tools and uses returned observations to choose the next action



Repeat until success or budget limit

Tools can keep separate, frozen weights

Yao et al. (2023) · ReAct / Wu et al. (2023) · Visual ChatGPT

ReAct Visual ChatGPT

Mug editing

“Move the red mug to the right of the blue plate, then explain the edit aloud”

Step	Model or tool call	Artifact returned to the controller
1 Locate	GroundingDINO: image + object names	Mug and plate bounding boxes
2 Isolate	SAM: image + mug box	Mug segmentation mask
3 Move + repair	Code: relocate masked pixels Stable Diffusion inpainting: fill + blend	Candidate image + edit mask Preserve pixels outside the mask
4 Check / retry	Qwen2.5-VL: original + candidate + request	Relation, identity, preservation check Pass or repair instruction
5 Explain aloud	TTS: description of the checked result	Spoken explanation

Wu et al. (2023) · Visual ChatGPT / Yao et al. (2023) · ReAct

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- How can we rule out the “just more parameters” in these MoE/MoT studies?